

SE LAB 1

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Problem statement - #30, Airport Lost Luggage Claim & Tracking Portal

Actors: Passenger, Baggage Service Agent

1.a. Functional Requirements Table

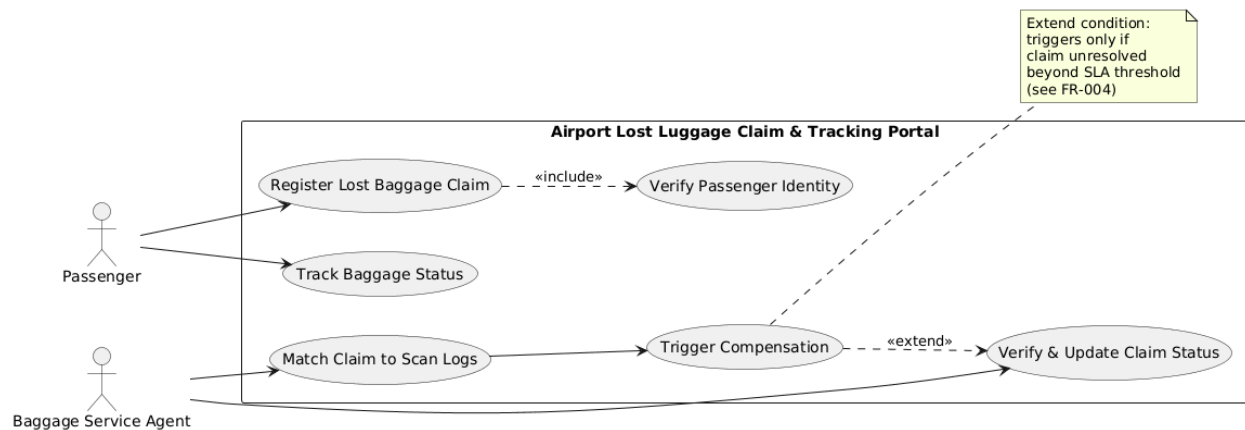
Id	Priority	Description	Acceptance Criteria	Rationale
FR-001	High	The system cross-references passenger lost baggage reports with unrouted luggage scan logs across all connected airport terminals.	Pass: If a match is identified and a recovery alert is generated. Fail: If an incorrect bag tag format is accepted.	Automated matching is the crux, manual cross-referencing across terminals is extremely slow.
FR-002	High	The system shall allow a Passenger to register a lost baggage claim by submitting flight details, bag tag ID, and a description of the bag	Pass: Claim created with unique claim ID and status "Submitted." Fail: Claim accepted without a valid flight/bag tag reference.	Passengers need a formal, trackable entry point before any recovery process can start.
FR-003	Medium	The system shall allow a Passenger to view the live status and scan-location timeline of their claim at any time.	Pass: Timeline reflects the latest scan event within the NFR-001 latency bound. Fail: Stale or missing scan data shown as current.	Reduces support-desk load and passenger anxiety by giving self-service visibility.
FR-004	High	The system shall automatically initiate a compensation workflow if a claim remains unresolved beyond a configurable SLA threshold (e.g., 96 hours).	Pass: Compensation case auto-created and Agent notified at threshold breach. Fail: No action taken after SLA lapses.	Airlines are contractually/regulatorily obligated to compensate delayed baggage; automation prevents missed deadlines.
FR-005	High	The system shall allow a Baggage Service Agent to verify claim details against physical recovery and update claim status (e.g., Matched, In Transit, Delivered, Closed).	Pass: Status change is logged with agent ID and timestamp. Fail: Status changes without an audit trail.	Agents are the human checkpoint ensuring system matches are correct before closing a claim.

1.b. Non-Functional Requirements Table

ID	Type	Description	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms.	Pass: Benchmarking tests confirm target latency and	Slow or insecure APIs undermine passenger

			security standards under simulated peak load.	trust and risk exposing PII during transit.
NFR-002	Availability	The baggage tracking portal shall maintain 99.9% uptime, excluding scheduled maintenance windows.	Pass: Uptime monitoring confirms ≥99.9% over a rolling 30-day window. Fail: Unplanned downtime exceeds SLA.	Passengers and agents need continuous access, especially during peak travel disruption events when baggage issues spike.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case Name: Match Claim to Scan Logs

Actors: Baggage Service Agent (primary), Passenger (secondary — receives outcome)

Preconditions:

- A lost baggage claim (FR-002) has been registered by the Passenger with a valid bag tag ID and flight details.
- The claim status is "Submitted."
- Bag tag scan logs from all connected airport terminals are available and up to date.

Main Success Scenario:

1. Agent (or scheduled system job) initiates a cross-reference check for a submitted claim.
2. System retrieves the claim's bag tag ID and flight/route details.
3. System queries unrouted luggage scan logs across all connected terminals for a matching bag tag ID.
4. System finds a scan record matching the bag tag ID and route.
5. System generates a recovery alert and links the scan record to the claim.
6. System updates the claim status to "Matched" and timestamps the match.

7. System notifies the Agent and the Passenger of the match, including the bag's current terminal location.
8. Agent confirms the match against physical recovery and updates the claim status to "In Transit" or "Delivered" (FR-005).

Alternate Flow — No Match Found:

- 3a. System completes the scan log query but finds no matching bag tag ID.
- 3b. System sets claim status to "Unresolved" and schedules an automatic re-check (e.g., every 6 hours).
- 3c. If the claim remains "Unresolved" beyond the SLA threshold, the system triggers the compensation workflow (FR-004) and notifies the Agent for possible escalation.
- 3d. Flow rejoins at step 7 once a later re-check produces a match, or ends if compensation is issued and the claim is closed without recovery.

Postconditions:

- Claim status is updated to reflect the outcome ("Matched," "Unresolved," or escalated to compensation).
- An audit trail entry is created recording the match attempt, timestamp, and responsible actor (system/Agent).
- Passenger has visibility into the updated status via the tracking view (FR-003).